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-->

CVarArg

A type whose instances can be encoded, and appropriately passed, as elements of a C `va_list`.

```
protocol CVarArg
```

Overview

You use this protocol to present a native Swift interface to a C “varargs” API. For example, a program can import a C API like the one defined here:

```c
int c_api(int, va_list arguments)
```

To create a wrapper for the `c_api` function, write a function that takes `CVarArg` arguments, and then call the imported C function using the `withVaList(_:_:)` function:

```
func swiftAPI(_ x: Int, arguments: CVarArg...) -> Int {
    return withVaList(arguments) { c_api(x, $0) }
}
```

Swift only imports C variadic functions that use a `va_list` for their arguments. C functions that use the `...` syntax for variadic arguments are not imported, and therefore can’t be called using `CVarArg` arguments.

If you need to pass an optional pointer as a `CVarArg` argument, use the `Int(bitPattern:)` initializer to interpret the optional pointer as an `Int` value, which has the same C variadic calling conventions as a pointer

on all supported platforms.

> Note: Declaring conformance to the `CVarArg` protocol for types defined
> outside the standard library is not supported.

Relationships

Conforming Types

[`Bool`](/documentation/Swift/Bool)

[`Dictionary`](/documentation/Swift/Dictionary)

[`UnsafeMutablePointer`](/documentation/Swift/UnsafeMutablePointer)

[`UInt32`](/documentation/Swift/UInt32)

[`Int32`](/documentation/Swift/Int32)

[`Set`](/documentation/Swift/Set)

[`AutoreleasingUnsafeMutablePointer`](/documentation/Swift/AutoreleasingUnsafeMutablePointer)

[`Float`](/documentation/Swift/Float)

[`Int16`](/documentation/Swift/Int16)

[`OpaquePointer`](/documentation/Swift/OpaquePointer)

[`Int8`](/documentation/Swift/Int8)

[`UInt64`](/documentation/Swift/UInt64)

[`UInt16`](/documentation/Swift/UInt16)

[`UnsafePointer`](/documentation/Swift/UnsafePointer)

[`String`](/documentation/Swift/String)

[`UInt8`](/documentation/Swift/UInt8)

[`Int64`](/documentation/Swift/Int64)

[`Float80`](/documentation/Swift/Float80)

[`UInt`](/documentation/Swift/UInt)

[`Int`](/documentation/Swift/Int)

[`Double`](/documentation/Swift/Double)

[`Array`](/documentation/Swift/Array)
